The circuit consists of two parts: the power source and the LED flasher.

Power source:
In a bright environment, the solar cell produces a voltage, and if it is higher than the voltage in the storage capacitor, a current can flow from the solar cell into the storage capacitor. In darkness, the flasher circuit operates on the voltage available in the storage capacitor.
The diode prevents current from "flowing back" through the solar cell when the capacitor is charged but the voltage at the solar cell is low in the dark.

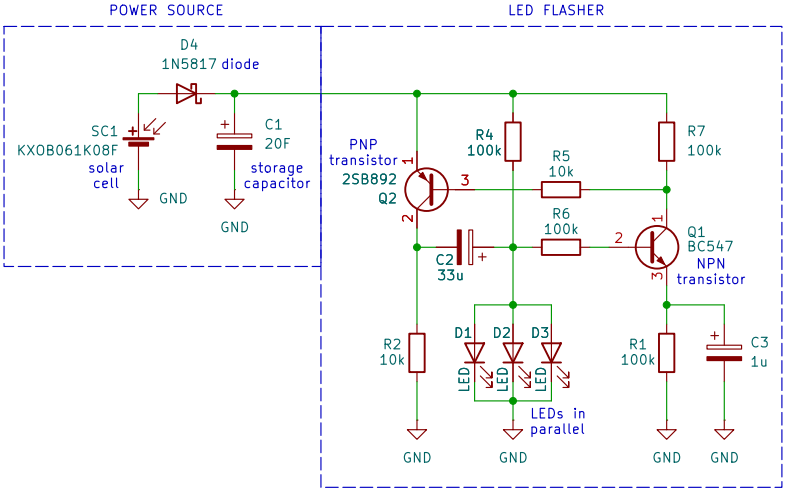
Solar cell:
ANYSOLAR makes beautiful, small solar cells, especially the IXOLAR SolarBIT (KXOB*) and SolarMD (SM*) series.
We use a cell with 5.53V nominal open-circuit voltage. Our experiments showed that the voltage after the diode at the main capacitor doesn't exceed the maximum 3.8V that our capacitor allows.

Storage capacitor:
We use an EDLC (Electrochemical Double-Layer Capacitor) with a maximum voltage of 3.8V. The voltage coming from the solar cell in bright light mustn't exceed the maximum voltage of the capacitor.

Transistors:
The transistors can be standard NPN and PNP types with high enough gain.
As NPN, we use a BC547, maybe the most common NPN type in Europe.
The US equivalent would be the 2N3904, which should work fine here.
As PNP, we use the more exotic 2SB892, simply because of its elongated body shape so it can be easily visually distinguished in the DIY kit.
A BC557 or 2N3906 should work fine here.

LEDs:
We use red LEDs with a low forward voltage (Vf) of about 1.8V. Green and yellow LEDs with a Vf up to about 2.2V might possibly work; LEDs with an even higher Vf (blue, white) will most probably not work.
The three LEDs are in parallel; one or two LEDs would flash a bit brighter, more LEDs in parallel less bright.

Other components:
The values of the other resistors and capacitors are carefully chosen and form a quite complex system. Values can be modified; e.g., changing the capacitance of the 1µF capacitor will affect the length of the flashes, and changing the 33µF capacitor, the time between flashes.
But the components of the whole flasher circuit form a sensitive system, and changing the values of some components requires changing the values of some other components to maintain this delicate balance. Smaller changes might work, but if some values are changed too much, the whole circuit stops working.



schematic design: Steffen Hartwig
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Title: Raumgleiter (LEGIBLE SCHEMATIC)

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